

Curriculum vitae

* ROLE IN THE PROJECT

Project manager ☐ Work package leader ☐
Project partner ☒ Other (specify) ☐

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* PERSONAL INFORMATION

*Family name, First name:	Kording, Konrad P		
*Date of birth:	1973	*Sex:	Male
*Nationality:	United States Citizen/German		

* HIGHER EDUCATION/OTHER TRAINING

	Subjects/degree/	Name of institution, country
2001	Neuroinformatics / PhD	ETH Zurich, Switzerland
1997	Physics / BSc	ETH Zurich, Switzerland

* **POSITIONS** (academic, business, industry, public sector, national or international organisations)

Current Position

	Job title/name of employer/country
2017-	Professor / University of Pennsylvania / USA

Previous positions held (list)

	Job title/name of employer/country
2015-2017	Full Professor / Northwestern University / USA
2011-2015	Associate Professor / Northwestern University, PM&R, Physiology, Applied Math and at the Rehabilitation Institute of Chicago / USA
2006-2011	Assistant Professor, Northwestern University / PM&R, Physiology, Applied Math and at the Rehabilitation Institute of Chicago / USA
2004-2006	Heisenberg fellow with Josh Tenenbaum, Machine Learning
2002-2004	Postdoc, addressing learning and optimality in the motor system with Daniel Wolpert
2001-2002	Postdoc, Collegium Helveticum

PROJECT MANAGEMENT EXPERIENCE (if applicable)

	Project/topic/role in project/funding from
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2009-2020	R01NS063399 The Role of Uncertainty in Human Motor Learning and Adaptation /Principal Investigator / The National Institutes of Health, USA
2013-2019	R01MH103910 Recording Neural Activities onto DNA /Principal Investigator / The National Institutes of Health, USA.
2015-2019	U01NS094248 Massive scale electrical neural recordings in vivo using commercial ROIC chips / Principal Investigator / The National Institutes of Health, USA.
2017-2021	DARPA-BAA-16-09 NESD IBIS: Implantable bioluminescence interface system for an all-optical neuroprosthesis to the visual cortex / Principal Investigator/ Neural Engineering System Design (NESD), USA
2017-2021	1R01MH111610-01A1LifeSense: Transforming Behavioral Assessment of Depression Using Personal Sensing Technology /Principal Investigator /The National Institutes of Health, USA.
2017-2020	NIH R01EY021579 Neural Mechanisms of Fixation Choice while Searching Natural Scenes / Principal Investigator / The National Institutes of Health, USA.
2018-2020	NIH R24MH114799 SABER: Scalable Analytics for Brain Exploration Research using X-Ray Microtomography and Electron Microscopy / Principal Investigator / The National Institutes of Health, USA.

OTHER MERITS RELEVANT TO THE PROJECT

Contributions to Science

1. Movement science The central idea behind Bayesian models of behavior is that the brain solves a statistical estimation problem to deal with noisy and ambiguous sensors and muscles. In Bayesian modeling, we first define the statistical problem that the brain has to solve. We then calculate the optimal statistical solution to the problem. Lastly, we compare this statistical solution to the way the brain does solve the problem. For the kind of sensorimotor behaviors that I study, the brain is strikingly good at solving the underlying statistical problems. My lab has spearheaded this approach in the domain of motor control:

- a) Berniker M, Kording KP (2008). Motor Adaptation: Estimating the sources of errors. *Nature Neuroscience* 11: 1454-1461. PMID: PMC2707921
- b) Kording KP (2007). Decision theory: what “should” the nervous system do? *Science* 26, 318(5850): 606-610.
- c) Kording KP, Wolpert D (2004). Bayesian Integration in Sensorimotor Learning. *Nature* 427(6971): 244-247.

2. Computational principles in neurophysiology The question of how the brain *mechanistically* implements its computations is one of the central questions in neuroscience. Only careful physiological experiments combined with modeling efforts can provide an understanding of the way the brain solves its computational problems. My lab uses physiological approaches in collaboration with other experimentalists using humans, monkeys, and rodents as model system. It then uses state of the art data analysis techniques to analyze the results.

- a) Fernandes HL, Stevenson IH, Phillips AN, Segraves MA, Kording KP (2013) Saliency and Saccade Encoding in the Frontal Eye Field During Natural Scene Search. *Cereb Cortex*. PMID: PMC4240184
- b) Rebesco, JM, Stevenson, IH, Kording, KP, Solla, SA, & Miller, LE (2010). Rewiring neural interactions by micro-stimulation. *Frontiers in systems neuroscience*, 4. PMID: PMC2936935

- c) Glaser JI, Wood DK, Lawlor PN, Ramkumar P, Kording KP, Segraves MA (2016) The role of expected reward in frontal eye field during natural scene search. *J Neurophysiol*:jn 00119 02016.
- d) Ramkumar P, Lawlor PN, Glaser JI, Wood DK, Phillips AN, Segraves MA, Kording KP (2016) Feature-based attention and spatial selection in frontal eye fields during natural scene search. *J Neurophysiol*:jn 01044 02015.

3. Neural data analysis Neuroscience is producing some of the biggest datasets in science and a central problem is describing the data in terms of principles. We have focused extensively on developing neural data analysis algorithms that scale well with data size and can thus take advantage of the exponential growth in datasets. Of particular interest is transforming recorded spike data into insights into the way neurons interact. Much of our focus has been on using Bayesian methods to allow reliable inferences about many variables at the same time.

- a) Stevenson IH, Kording KP (2011). How advances in neural recording affect data analysis. *Nature Neuroscience* 14: 139-142. PMID: PMC3410539
- b) Stevenson IH, Cherian A, London BM, Sachs NA, Lindberg E, Reimer J, Slutzky MW, Hatsopoulos NG, Miller LE, Kording KP (2011). Statistical assessment of the stability of neural movement representations. *J. Neurophysiol.* 106(2): 764-774. PMID: PMC3154833
- c) Stevenson IH, London BM, Oby ER, Sachs NA, Reimer J, Englitz B, David SV, Shamma SA, Blanche TJ, Mizuseki K, Zandvakili A, Hatsopoulos NG, Miller LE, Kording KP (2012). Functional Connectivity and Tuning Curves in Populations of Simultaneously Recorded Neurons. *PLoS Comput. Biol.* 8(11): e1002775. PMID: PMC3499254

4. Medical machine learning The ubiquity of sensors promises making clinically relevant diagnoses based on everyday data. As these approaches are thus promising to scale towards a world-wide application, they are very important. At the same time, the field has so far used rather low standards. My work thus distributes between applications work and meta-work that is aimed at improving the standards used in medical machine learning.

- a) Saeb S, Lonini K, Jayaraman A, Mohr DC, Kording KP. (2017) The need to approximate the use-case in clinical machine learning, *GigaScience* 6 (5), 1-9
- b) DeMasi O, Kording KP, Recht B (2017) Meaningless comparisons lead to false optimism in medical machine learning, *PloS one* 12 (9), e0184604
- c) Saeb S, Zhang M, Karr CJ, Schueller SM, Corden ME, Kording KP, Mohr D (2016) Mobile phone sensor correlates of depressive symptom severity in daily-life behavior: an exploratory study, *Journal of medical Internet research* 17 (7)
- d) Albert MV, Kording KP, Herrmann M, Jayaraman A (2015) Fall classification by machine learning using mobile phones, *PloS one* 7 (5), e36556

Complete list of published work:

<https://scholar.google.com/citations?user=MIFqJGcAAAAJ&hl=en>